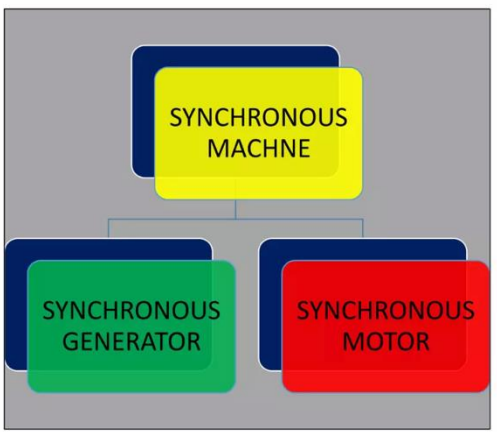
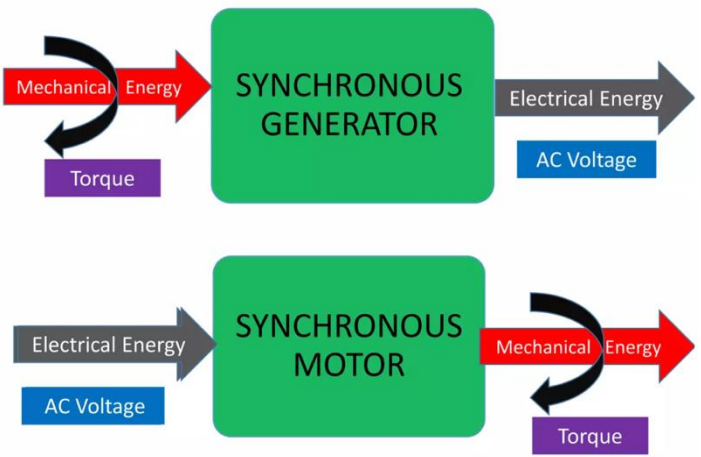
Synchronous Machines



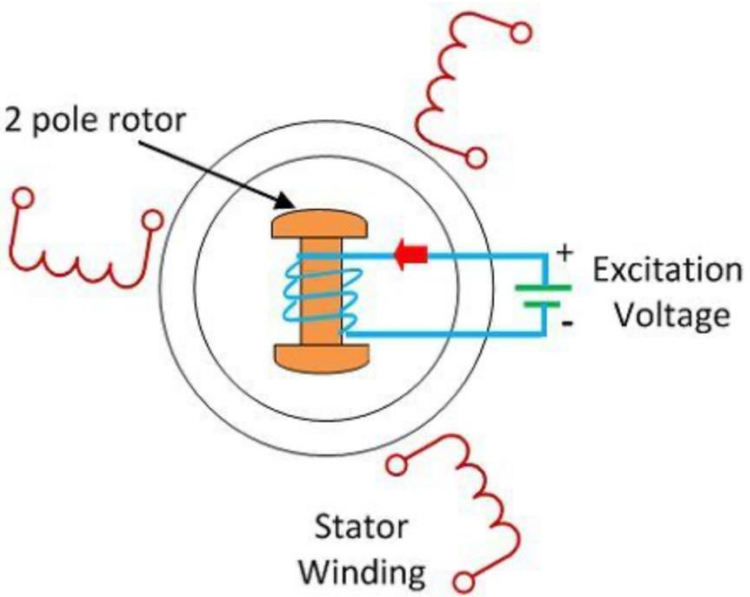


Operation Principle and Construction of a Synchronous Generator

- **Synchronous generators** are by definition ***synchronous***, meaning that the electrical frequency produced in the stator is ***locked*** or ***synchronized*** with the mechanical rate of rotation of the generator.

- In another words, the speed of rotating magnetic field and mechanical rate of rotation are **equal.**

Operation Principle and Construction of a Synchronous Generator



Synchronous Generator

Operation Principle and Construction of a Synchronous Generator

Synchronous machines are AC machines that have a field circuit supplied by an external DC source.

In a synchronous generator, a DC current is applied to the rotor winding producing a rotor magnetic field. The rotor is then turned by external means producing a rotating magnetic field, which induces a 3-phase voltage within the stator winding.

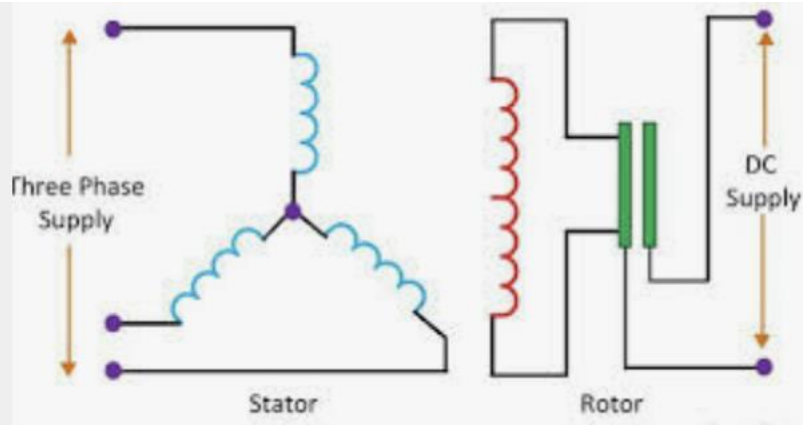
In synchronous generator

- **Armature windings on Stator**
- **Field windings on Rotor – Which is energized by dc source to develop rotor magnetic field.**
- **Rotor is turned by a prime mover, producing a rotating magnetic field which induce 3 phase voltages in stator windings**

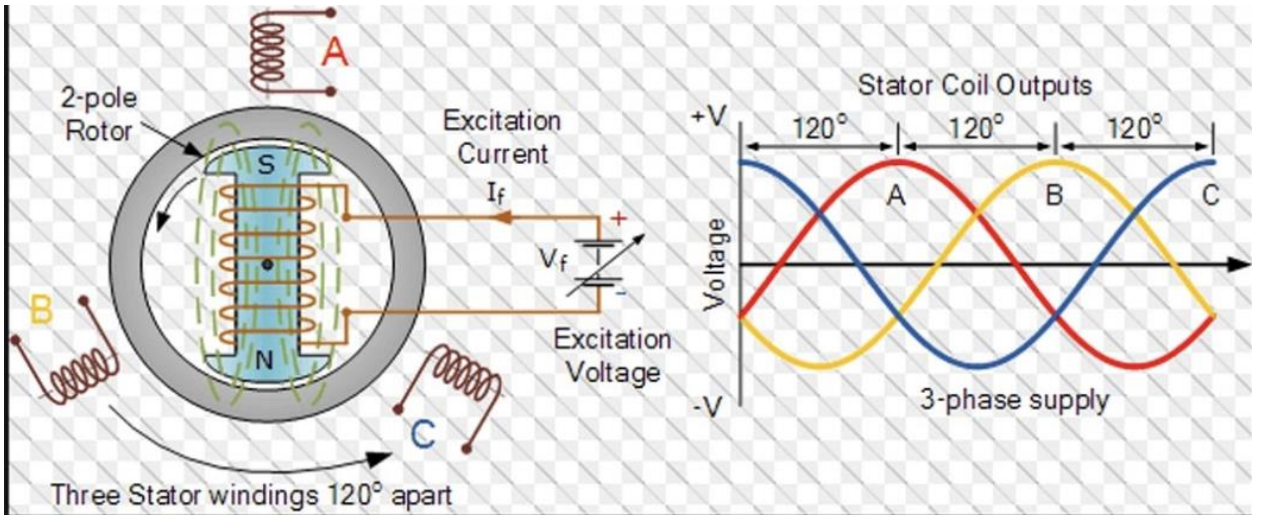
Operation Principle and Construction of a Synchronous Generator

- In a synchronous generator, the rotor winding (field winding) produces the rotating magnetic field.
- This rotating magnetic field induces a voltage in the stator winding, leading to the generation of a magnetic field in the stator.
- The interaction between the rotor and stator magnetic fields establishes the torque necessary for the generator to produce electrical power.

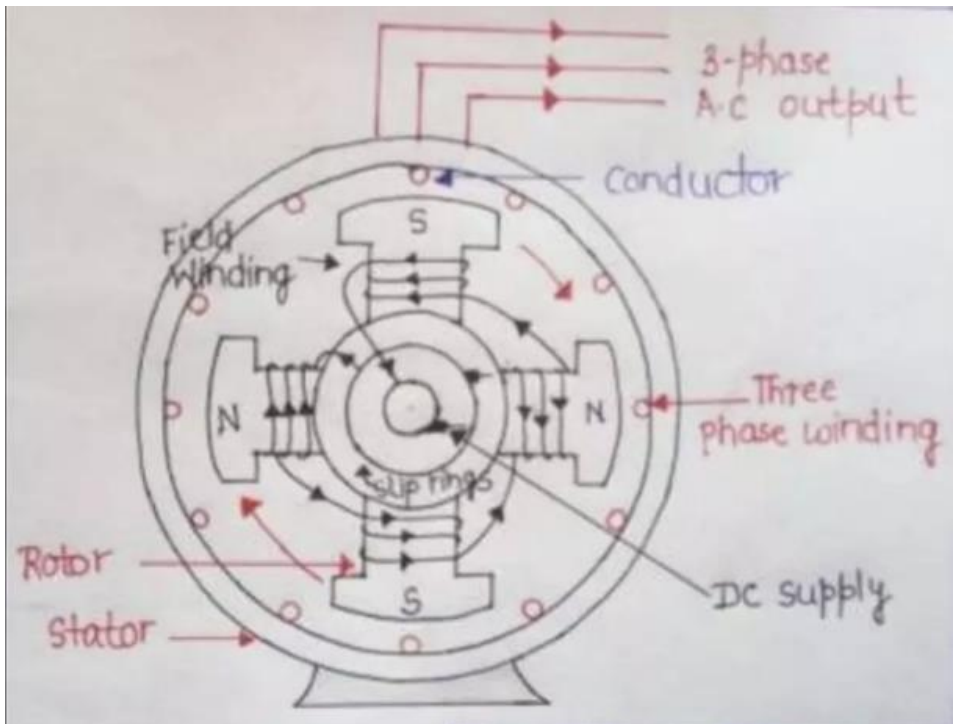
Operation Principle and Construction of a Synchronous Generator



Operation Principle and Construction of a Synchronous Generator

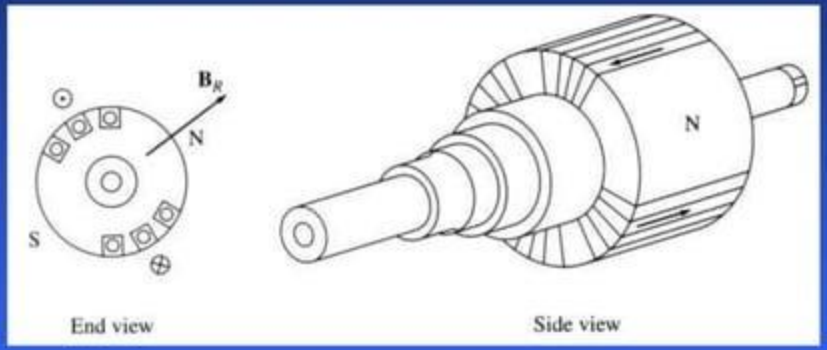


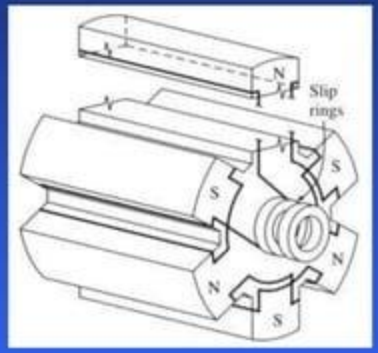
Operation Principle and Construction of a Synchronous Generator



Construction of Synchronous Machines

The rotor of a synchronous machine is a large electromagnet. The magnetic poles can be either salient (sticking out of rotor surface) or non-salient construction.





Non-salient-pole rotor: usually two- and four-pole rotors.

Salient-pole rotor: four and more poles.

Rotors are made laminated to reduce eddy current losses.

Construction of Synchronous Machines

Two common approaches are used to supply a DC current to the field circuits on the rotating rotor:

1. Supply the DC power from an external DC source to the rotor by means of slip rings and brushes;
2. Supply the DC power from a special DC power source mounted directly on the shaft of the machine.



Slip rings are metal rings completely encircling the shaft of a machine but insulated from it. One end of a DC rotor winding is connected to each of the two slip rings on the machine's shaft. Graphite-like carbon brushes connected to DC terminals ride on each slip ring supplying DC voltage to field windings regardless the position or speed of the rotor.

Rotation speed of Synchronous Generator

By the definition, synchronous generators produce electricity whose frequency is synchronized with the mechanical rotational speed.

$$f_e = \frac{n_m P}{120} \qquad (7.11.1)$$

Where f_e is the electrical frequency, Hz;
n_m is mechanical speed of magnetic field (rotor speed for synchronous machine), rpm;
P is the number of poles.

Steam turbines are most efficient when rotating at high speed; therefore, to generate 60 Hz, they are usually rotating at 3600 rpm and turn 2-pole generators.

Water turbines are most efficient when rotating at low speeds (200-300 rpm); therefore, they usually turn generators with many poles.

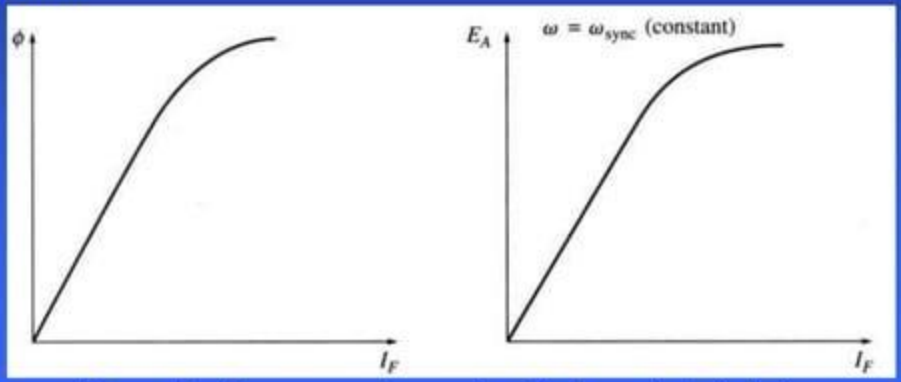
Internal generated voltage of a Synchronous Generator

The magnitude of internal generated voltage induced in a given stator is

$$E_A = \sqrt{2}\pi N_C \phi f = K\phi\omega$$

where K is a constant representing the construction of the machine, ϕ is flux in it and ω is its rotation speed.

Since flux in the machine depends on the field current through it, the internal generated voltage is a function of the rotor field current.



Magnetization curve (open-circuit characteristic) of a synchronous machine

Equivalent circuit of a Synchronous Generator

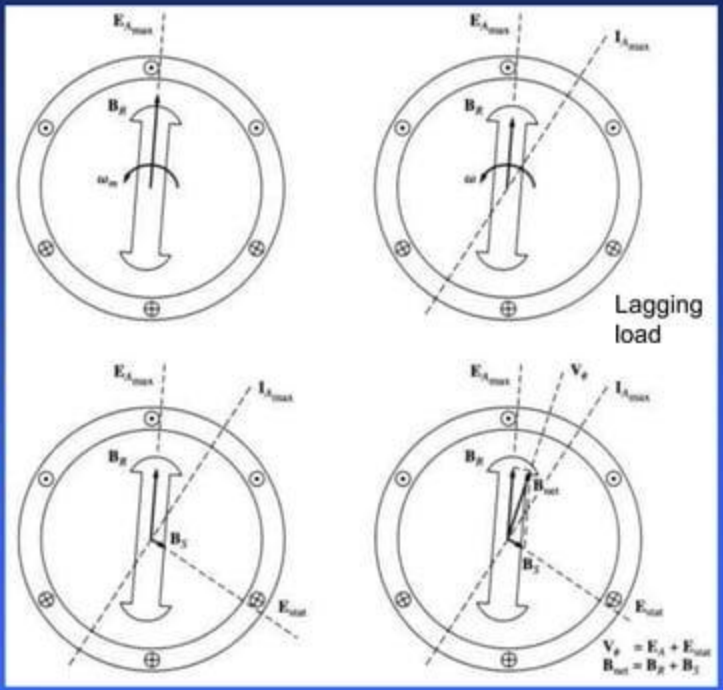
The internally generated voltage in a single phase of a synchronous machine E_A is not usually the voltage appearing at its terminals. It equals to the output voltage V_ϕ only when there is no armature current in the machine. The reasons that the armature voltage E_A is not equal to the output voltage V_ϕ are:

1. Distortion of the air-gap magnetic field caused by the current flowing in the stator (armature reaction);
2. Self-inductance of the armature coils;
3. Resistance of the armature coils;
4. Effect of salient-pole rotor shapes.

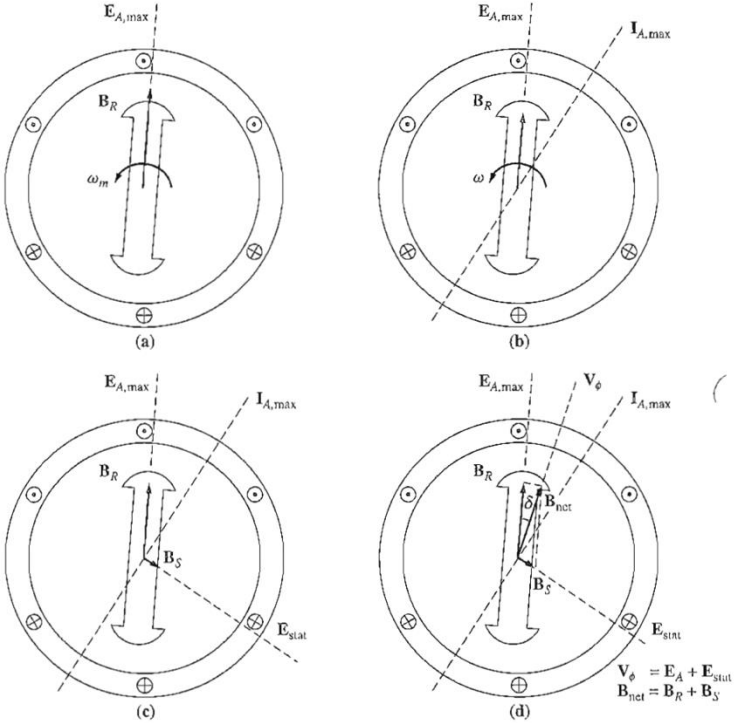
Equivalent Circuit of a Synchronous Generator

Armature reaction (the largest effect):

When the rotor of a synchronous generator is spinning, a voltage E_A is induced in its stator. When a load is connected, a current starts flowing creating a magnetic field in machine's stator. This stator magnetic field B_S adds to the rotor (main) magnetic field B_R affecting the total magnetic field and, therefore, the phase voltage.



Effect of armature reaction



The development of a model for armature reaction: (a) A rotating magnetic field produces the internal generated voltage $\mathbf{E}_A$. (b) The resulting voltage produces a lagging *current flow* when connected to a lagging load. (c) The stator current produces its own magnetic field $\mathbf{B}_S$, which produces its own voltage $\mathbf{E}_{stat}$ in the stator windings of the machine. (d) The field $\mathbf{B}_S$ adds to $\mathbf{B}_R$, distorting it into $\mathbf{B}_{net}$. The voltage $\mathbf{E}_{stat}$ adds to $\mathbf{E}_A$, producing $\mathbf{V}_\phi$ at the output of the phase.

Equivalent circuit of a Synchronous Generator

Assuming that the generator is connected to a lagging load, the load current I_A will create a stator magnetic field B_S, which will produce the armature reaction voltage E_{stat}. Therefore, the phase voltage will be

$$V_\phi = E_A + E_{stat} \qquad (7.16.1$$

The net magnetic flux will be

$$B_{net} = \underset{\text{Rotor field}}{B_R} + \underset{\text{Stator field}}{B_S} \qquad (7.16.2$$

Note that the directions of the net magnetic flux and the phase voltage are the same.

Equivalent circuit of a Synchronous Generator

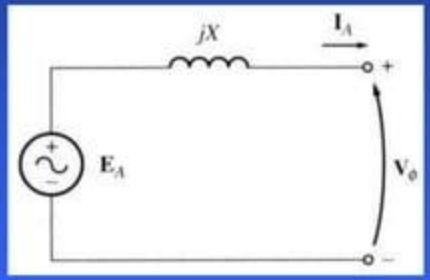
Assuming that the ~~load~~ armature reactance is X, the armature reaction voltage is

$$E_{stat} = -jXI_A \qquad (7.17.1)$$

The phase voltage is then

$$V_\phi = E_A - jXI_A \qquad (7.17.2)$$

Armature reactance can be modeled by the following circuit…



However, in addition to armature ~~reactance~~ reaction effect, the stator coil has a self-inductance L_A (X_A is the corresponding reactance) and the stator has resistance R_A. The phase voltage is thus

$$V_\phi = E_A - jXI_A - jX_AI_A - RI_A \qquad (7.17.3)$$

V_{AR} $V_{stator\ winding}$

Equivalent circuit of a Synchronous Generator

Often, armature reactance and self-inductance are combined into the synchronous reactance of the machine:

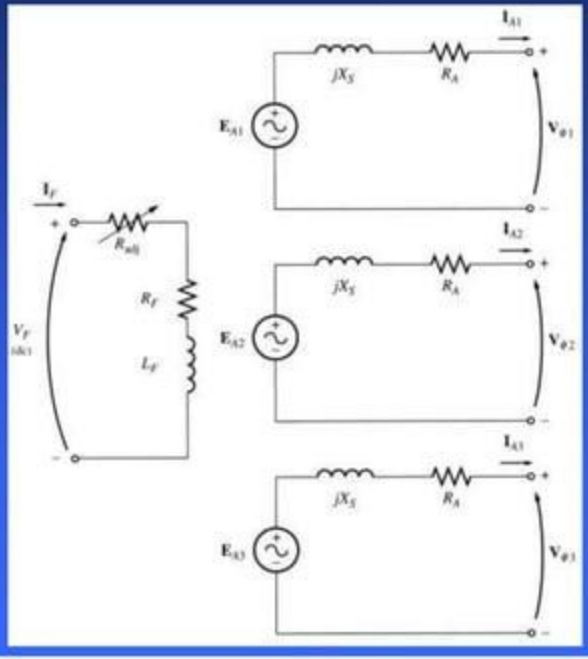
$$X_S = X + X_A \qquad (7.18.1)$$

Therefore, the phase voltage is

$$V_\phi = E_A - jX_S I_A - RI_A \qquad (7.18.2)$$

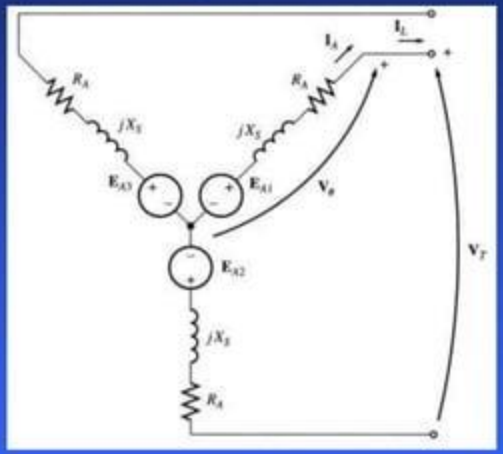
The equivalent circuit of a 3-phase synchronous generator is shown.

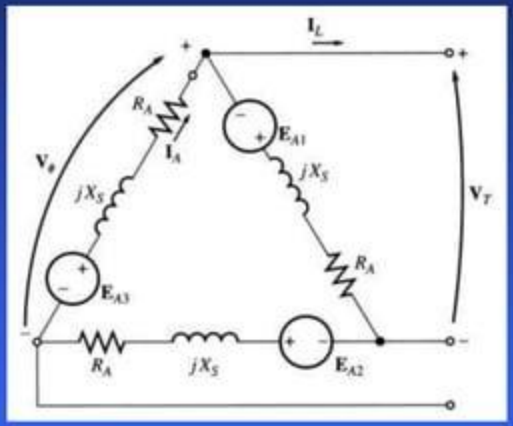
The adjustable resistor R_{adj} controls the field current and, therefore, the rotor magnetic field.



Equivalent circuit of a Synchronous Generator

A synchronous generator can be Y- or Δ-connected:





The terminal voltage will be

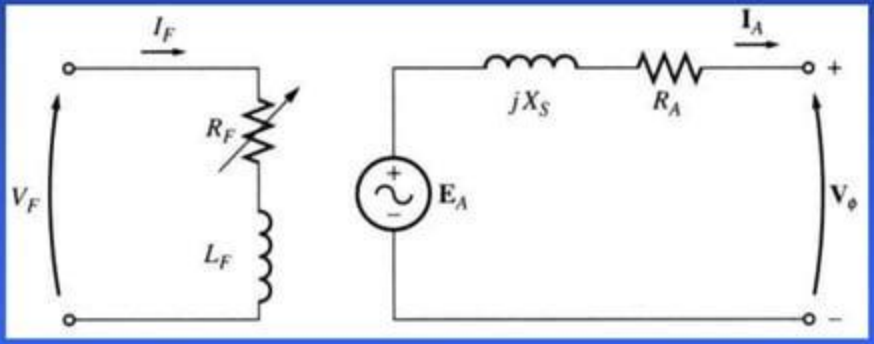
$$V_T = \sqrt{3}V_\phi \rightarrow for\ Y \qquad (7.19.1)$$

$$V_T = V_\phi \rightarrow for\ \Delta \qquad (7.19.2)$$

Equivalent circuit of a synchronous generator

Note: the discussion above assumed a balanced load on the generator!

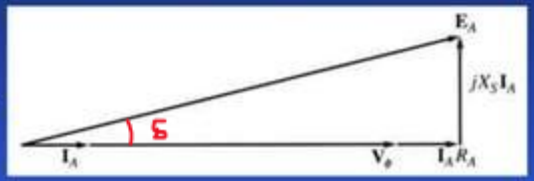
Since – for balanced loads – the three phases of a synchronous generator are identical except for phase angles, per-phase equivalent circuits are often used.



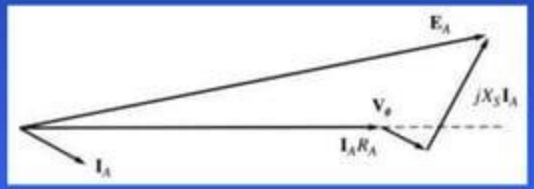
Phasor diagram of a Synchronous Generator

Since the voltages in a synchronous generator are AC voltages, they are usually expressed as phasors. A vector plot of voltages and currents within one phase is called a phasor diagram.

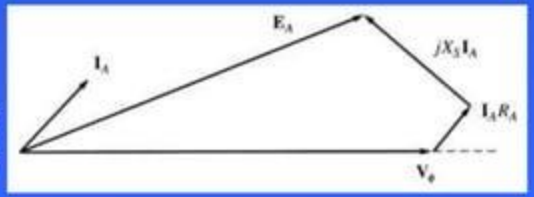
A phasor diagram of a synchronous generator with a unity power factor (resistive load) ⟶



Lagging power factor (inductive load): a larger than for leading PF internal generated voltage E_A is needed to form the same phase voltage.



Leading power factor (capacitive load).



For a given field current and magnitude of load current, the terminal voltage is lower for lagging loads and higher for leading loads.

Power & torque in Synchronous Generators

A synchronous generator needs to be connected to a prime mover whose speed is reasonably constant (to ensure constant frequency of the generated voltage) for various loads.
The applied mechanical power

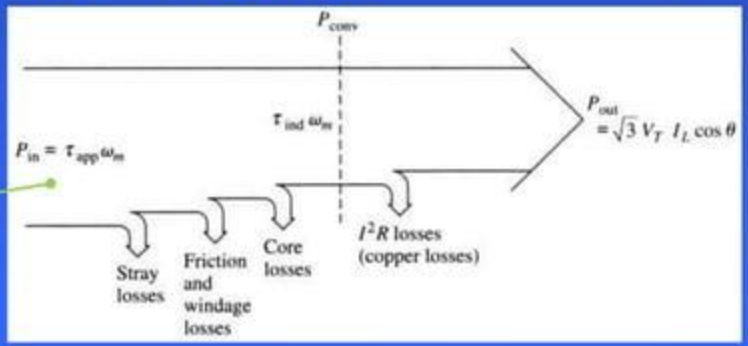
$$P_{in} = \tau_{app}\omega_m \tag{7.22.1}$$

is partially converted to electricity

$$P_{conv} = \tau_{ind}\omega_m = 3E_A I_A \cos\gamma \tag{7.22.2}$$

Where γ is the angle between E_A and I_A.

The power-flow diagram of a synchronous generator.



Power & torque in Synchronous Generators

The real output power of the synchronous generator is

$$P_{out} = \sqrt{3} V_T I_L \cos\theta = 3 V_\phi I_A \cos\theta \qquad (7.23.1)$$

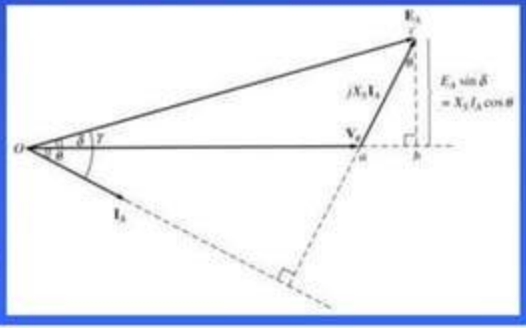
The reactive output power of the synchronous generator is

$$Q_{out} = \sqrt{3} V_T I_L \sin\theta = 3 V_\phi I_A \sin\theta \qquad (7.23.2)$$

Recall that the power factor angle θ is the angle between V_ϕ and I_A and **not** the angle between V_T and I_L.

In real synchronous machines of any size, the armature resistance $R_A << X_S$ and, therefore, the armature resistance can be ignored. Thus, a simplified phasor diagram indicates that

$$I_A \cos\theta = \frac{E_A \sin\delta}{X_S} \qquad (7.23.3)$$



Power & torque in Synchronous Generators

Then the real output power of the synchronous generator can be approximated as

$$P_{out} \approx \frac{3V_\phi E_A \sin\delta}{X_S} \qquad (7.24.1)$$

We observe that electrical losses are assumed to be zero since the resistance is neglected. Therefore:

$$P_{conv} \approx P_{out} \qquad (7.24.2)$$

Here δ is the torque angle of the machine – the angle between V_ϕ and E_A.

The maximum power can be supplied by the generator when $\delta = 90^0$:

$$P_{\max} = \frac{3V_\phi E_A}{X_S} \qquad (7.24.3)$$

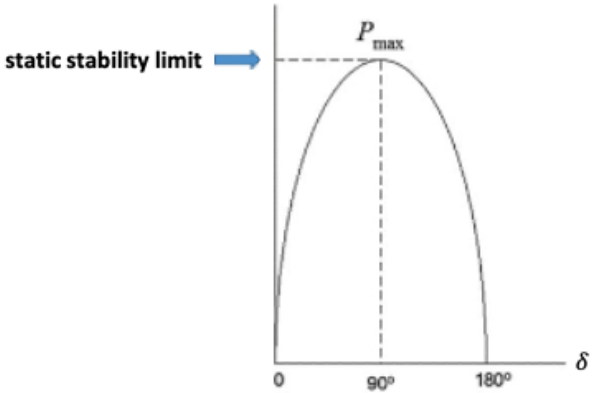
Power-flow diagram of a synchronous generator

- **Torque (power) angle** δ is increased by increasing driving torque
- **Maximum real power output** from a synchronous generator is attained **when** δ **= 90°**
- **If** δ **> 90°**, real power output suddenly drops to zero and **the machine losses synchronism**
- $P_{out(max)}$ is called "**steady-state stability limit**" of the synchronous generator
- Generally, synchronous generators are operated with a **small torque angle** δ **(15°**$\leq \boldsymbol{\delta} \leq$ **20°)**

$$P_{out} = 3V_{\emptyset}\frac{E_A sin\delta}{X_S} \quad \text{(if } \boldsymbol{R_A} \text{ is ignored)}$$

$$P_{out(max)} = \frac{3V_{\emptyset}E_A}{X_S} \quad \text{(when } \delta = 90°\text{)}$$

Real power output and **output electrical frequency** are controlled by **generator governor** that controls input mechanical power



Power & torque in synchronous generators

The maximum power specified by (7.24.3) is called the static stability limit of the generator. Normally, real generators do not approach this limit: full-load torque angles are usually between 15^0 and 20^0.

The induced torque is

$$\tau_{ind} = kB_R \times B_S = kB_R \times B_{net} = kB_R B_{net} \sin\delta \qquad (7.25.1)$$

Notice that the torque angle δ is also the angle between the rotor magnetic field B_R and the net magnetic field B_{net}.

Alternatively, the induced torque is

$$\tau_{ind} = \frac{3V_\phi E_A \sin\delta}{\omega_m X_S} \qquad (7.25.2)$$

Measuring parameters of Synchronous Generator Model

The three quantities must be determined in order to describe the generator model:

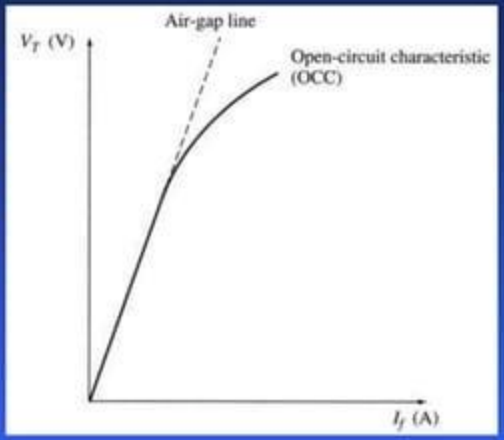
1. The relationship between field current and flux (and therefore between the field current I_F and the internal generated voltage E_A);
2. The synchronous reactance;
3. The armature resistance.

We conduct first the open-circuit test on the synchronous generator: the generator is rotated at the rated speed, all the terminals are disconnected from loads, the field current is set to zero first. Next, the field current is increased in steps and the phase voltage (which is equal to the internal generated voltage E_A since the armature current is zero) is measured.

Therefore, it is possible to plot the dependence of the internal generated voltage on the field current – the open-circuit characteristic (OCC) of the generator.

Measuring parameters of Synchronous Generator model

Since the unsaturated core of the machine has a reluctance thousands times lower than the reluctance of the air-gap, the resulting flux increases linearly first. When the saturation is reached, the core reluctance greatly increases causing the flux to increase much slower with the increase of the mmf.



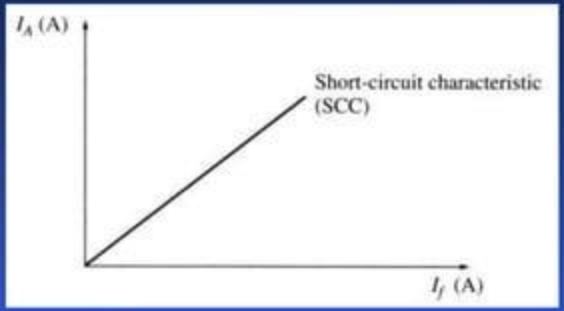
We conduct next the short-circuit test on the synchronous generator: the generator is rotated at the rated speed, all the terminals are short-circuited through ammeters, the field current is set to zero first. Next, the field current is increased in steps and the armature current I_A is measured as the field current is increased.

The plot of armature current (or line current) vs. the field current is the short-circuit characteristic (SCC) of the generator.

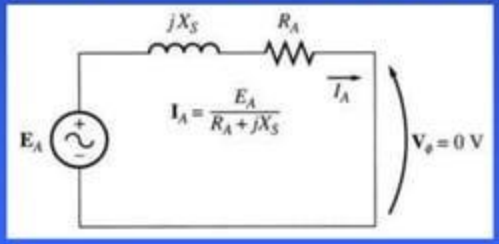
Measuring parameters of Synchronous Generator model

The SCC is a straight line since, for the short-circuited terminals, the magnitude of the armature current is

$$I_A = \frac{E_A}{\sqrt{R_A^2 + X_S^2}} \qquad (7.28.1)$$

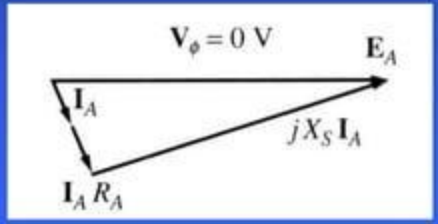


The equivalent generator's circuit during SC



Since B_S almost cancels B_R, the net field B_{net} is very small.

The resulting phasor diagram



The magnetic fields during short-circuit test



Measuring parameters of Synchronous Generator model

An approximate method to determine the synchronous reactance X_S at a given field current:

1. Get the internal generated voltage E_A from the OCC at that field current.
2. Get the short-circuit current $I_{A,SC}$ at that field current from the SCC.
3. Find X_S from

$$X_S \approx \frac{E_A}{I_{A,SC}} \tag{7.29.1}$$

Since the internal machine impedance is

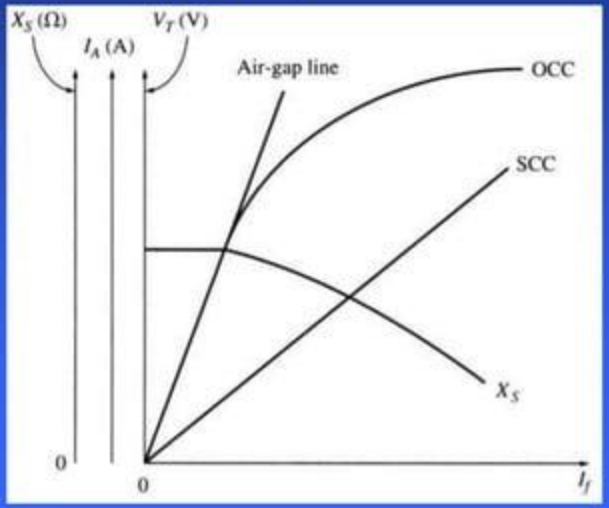
$$Z_S = \sqrt{R_A^2 + X_S^2} = \frac{E_A}{I_{A,SC}} \approx X_S \qquad \{\text{since } X_S \; ? \; R_A\} \tag{7.29.2}$$

Measuring parameters of Synchronous Generator model

A drawback of this method is that the internal generated voltage E_A is measured during the OCC, where the machine can be saturated for large field currents, while the armature current is measured in SCC, where the core is unsaturated. Therefore, this approach is accurate for unsaturated cores only.

The approximate value of synchronous reactance varies with the degree of saturation of the OCC.
Therefore, the value of the synchronous reactance for a given problem should be estimated at the approximate load of the machine.

The winding's resistance can be approximated by applying a DC voltage to a stationary machine's winding and measuring the current. However, AC resistance is slightly larger than DC resistance (skin effect).



Measuring parameters of Synchronous Generator model: Ex

Example 7.1: A 200 kVA, 480 V, 50 Hz, Y-connected synchronous generator with a rated field current of 5 A was tested and the following data were obtained:

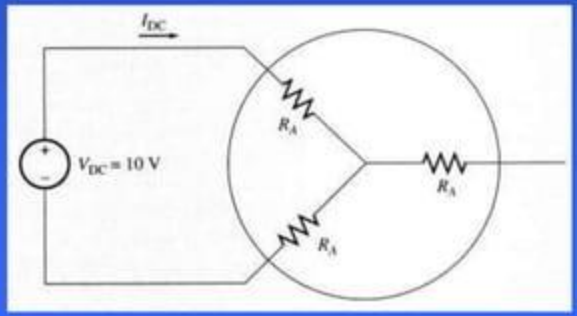
1. $V_{T,OC}$ = 540 V at the rated I_F.
2. $I_{L,SC}$ = 300 A at the rated I_F.
3. When a DC voltage of 10 V was applied to two of the terminals, a current of 25 A was measured.

Find the generator's model at the rated conditions (i.e., the armature resistance and the approximate synchronous reactance).

Since the generator is Y-connected, a DC voltage was applied between its ***two*** phases. Therefore:

$$2R_A = \frac{V_{DC}}{I_{DC}}$$

$$R_A = \frac{V_{DC}}{2I_{DC}} = \frac{10}{2 \cdot 25} = 0.2\,\Omega$$



Measuring parameters of Synchronous Generator model: Ex

The internal generated voltage at the rated field current is

$$E_A = V_{\phi,OC} = \frac{V_T}{\sqrt{3}} = \frac{540}{\sqrt{3}} = 311.8\,V$$

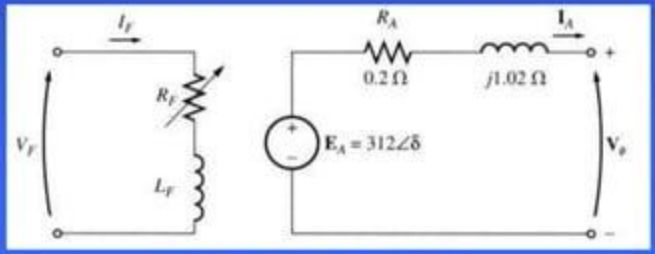
The synchronous reactance at the rated field current is precisely

$$X_S = \sqrt{Z_S^2 - R_A^2} = \sqrt{\frac{E_A^2}{I_{A,SC}^2} - R_A^2} = \sqrt{\frac{311.8^2}{300^2} - 0.2^2} = 1.02\,\Omega$$

We observe that if X_S was estimated via the approximate formula, the result would be:

$$X_S \approx \frac{E_A}{I_{A,SC}} = \frac{311.8}{300} = 1.04\,\Omega$$

Which is close to the previous result. The error ignoring R_A is much smaller than the error due to core saturation.



The equivalent circuit

The Synchronous Generator operating alone

The behavior of a synchronous generator varies greatly under load depending on the power factor of the load and on whether the generator is working alone or in parallel with other synchronous generators.

Although most of the synchronous generators in the world operate as parts of large power systems, we start our discussion assuming that the synchronous generator works alone.

Unless otherwise stated, the speed of the generator is assumed constant.

The Synchronous Generator operating alone

Effects of load changes



A increase in the load is an increase in the real and/or reactive power drawn from the generator.

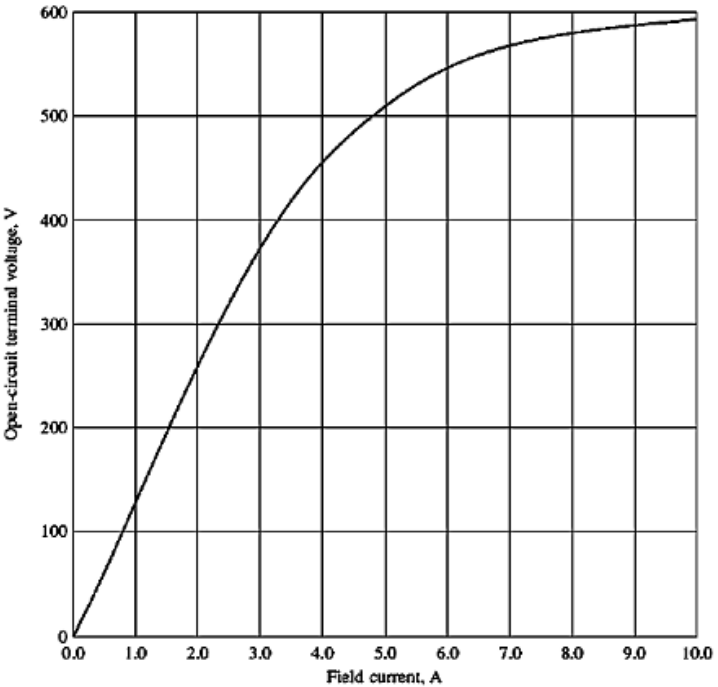
Since the field resistor is unaffected, the field current is constant and, therefore, the flux ϕ is constant too. Since the speed is assumed as constant, the magnitude of the internal generated voltage is constant also.

Assuming the same power factor of the load, change in load will change the magnitude of the armature current I_A. However, the angle will be the same (for a constant PF). Thus, the armature reaction voltage $jX_S I_A$ will be larger for the increased load. Since the magnitude of the internal generated voltage is constant

$$E_A = V_\phi + jX_S I_A \qquad (7.34.1)$$

Armature reaction voltage vector will "move parallel" to its initial position.

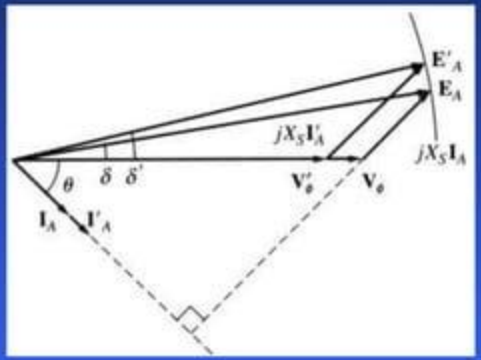
Synchronous generator operating alone



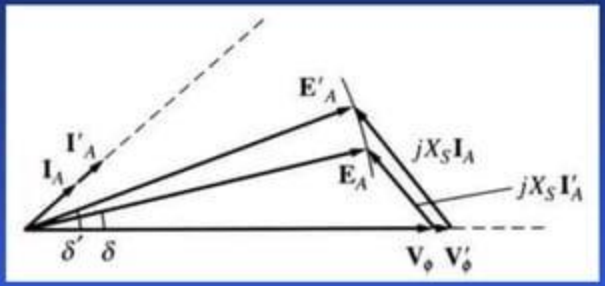
OCC characteristics of the synchronous generator

The Synchronous Generator operating alone

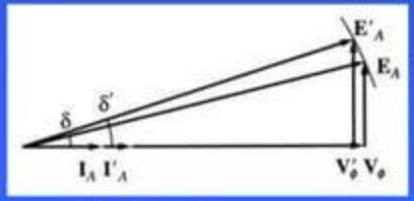
Increase load effect on generators with



Lagging PF



Leading PF



Unity PF

The Synchronous Generator operating alone

Generally, when a load on a synchronous generator is added, the following changes can be observed:

1. For lagging (inductive) loads, the phase (and terminal) voltage decreases significantly.
2. For unity power factor (purely resistive) loads, the phase (and terminal) voltage decreases slightly.
3. For leading (capacitive) loads, the phase (and terminal) voltage rises.

Effects of adding loads can be described by the voltage regulation:

$$VR = \frac{V_{nl} - V_{fl}}{V_{fl}} 100\% \qquad (7.36.1)$$

Where V_{nl} is the no-load voltage of the generator and V_{fl} is its full-load voltage.

The Synchronous Generator operating alone

A synchronous generator operating at a lagging power factor has a fairly large positive voltage regulation. A synchronous generator operating at a unity power factor has a small positive voltage regulation. A synchronous generator operating at a leading power factor often has a negative voltage regulation.

Normally, a constant terminal voltage supplied by a generator is desired. Since the armature reactance cannot be controlled, an obvious approach to adjust the terminal voltage is by controlling the internal generated voltage $E_A = K\phi\omega$. This may be done by changing flux in the machine while varying the value of the field resistance R_F, which is summarized:

1. Decreasing the field resistance increases the field current in the generator.
2. An increase in the field current increases the flux in the machine.
3. An increased flux leads to the increase in the internal generated voltage.
4. An increase in the internal generated voltage increases the terminal voltage of the generator.

Therefore, the terminal voltage of the generator can be changed by adjusting the field resistance.

The Synchronous Generator operating alone: Example

Example 7.2: A 480 V, 60 Hz, Y-connected six-pole synchronous generator has a per-phase synchronous reactance of 1.0 Ω. Its full-load armature current is 60 A at 0.8 PF lagging. Its friction and windage losses are 1.5 kW and core losses are 1.0 kW at 60 Hz at full load. Assume that the armature resistance (and, therefore, the I^2R losses) can be ignored. The field current has been adjusted such that the no-load terminal voltage is 480 V.

a. What is the speed of rotation of this generator?
b. What is the terminal voltage of the generator if
   1. It is loaded with the rated current at 0.8 PF lagging;
   2. It is loaded with the rated current at 1.0 PF;
   3. It is loaded with the rated current at 0.8 PF leading.
c. What is the efficiency of this generator (ignoring the unknown electrical losses) when it is operating at the rated current and 0.8 PF lagging?
d. How much shaft torque must be applied by the prime mover at the full load? how large is the induced countertorque?
e. What is the voltage regulation of this generator at 0.8 PF lagging? at 1.0 PF? at 0.8 PF leading?

The Synchronous Generator operating alone: Example

Since the generator is Y-connected, its phase voltage is

$$V_\phi = V_T / \sqrt{3} = 277\,V$$

At no load, the armature current I_A = 0 and the internal generated voltage is E_A = 277 V and it is constant since the field current was initially adjusted that way.

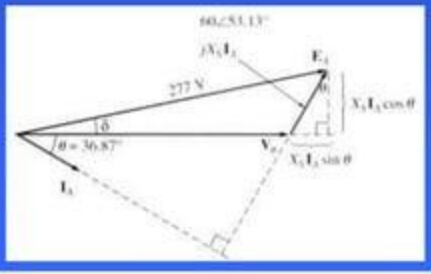
a. The speed of rotation of a synchronous generator is

$$n_m = \frac{120}{P} f_e = \frac{120}{6} 60 = 1200\,rpm$$

which is

$$\omega_m = \frac{1200}{60} 2\pi = 125.7\,rad/s$$

b.1. For the generator at the rated current and the 0.8 PF lagging, the phasor diagram is shown. The phase voltage is at 0^0, the magnitude of E_A is 277 V,



The Synchronous Generator operating alone: Example

and that $$jX_S I_A = j \cdot 1 \cdot 60\angle -36.87° = 60\angle 53.13°$$

Two unknown quantities are the magnitude of V_ϕ and the angle δ of E_A. From the phasor diagram:

$$E_A^2 = \left(V_\phi + X_S I_A \sin\theta\right)^2 + \left(X_S I_A \cos\theta\right)^2$$

Then:

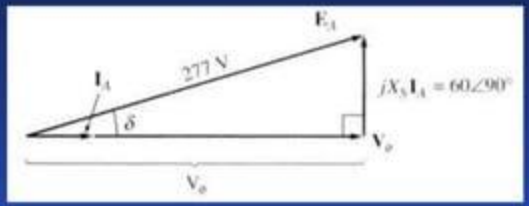
$$V_\phi = \sqrt{E_A^2 - \left(X_S I_A \cos\theta\right)^2} - X_S I_A \sin\theta = 236.8\,V$$

Since the generator is Y-connected,

$$V_T = \sqrt{3} V_\phi = 410\,V$$

The Synchronous Generator operating alone: Example

b.2. For the generator at the rated current and the 1.0 PF, the phasor diagram is shown. Then:

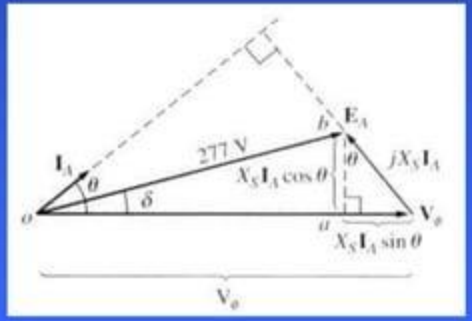


$$V_\phi = \sqrt{E_A^2 - (X_S I_A \cos\theta)^2} - X_S I_A \sin\theta = 270.4\,V$$

and

$$V_T = \sqrt{3}V_\phi = 468.4\,V$$

b.3. For the generator at the rated current and the 0.8 PF leading, the phasor diagram is shown. Then:



$$V_\phi = \sqrt{E_A^2 - (X_S I_A \cos\theta)^2} - X_S I_A \sin\theta = 308.8\,V$$

and

$$V_T = \sqrt{3}V_\phi = 535\,V$$

The Synchronous Generator operating alone: Example

c. The output power of the generator at 60 A and 0.8 PF lagging is

$$P_{out} = 3V_{\phi}I_A \cos\theta = 3 \cdot 236.8 \cdot 60 \cdot 0.8 = 34.1\,kW$$

The mechanical input power is given by

$$P_{in} = P_{out} + P_{elec\,loss} + P_{core\,loss} + P_{mech\,loss} = 34.1 + 0 + 1.0 + 1.5 = 36.6\,kW$$

The efficiency is

$$\eta = \frac{P_{out}}{P_{in}} \cdot 100\,\% = \frac{34.1}{36.6} \cdot 100\% = 93.2\%$$

d. The input torque of the generator is

$$\tau_{app} = \frac{P_{in}}{\omega_m} = \frac{36.6}{125.7} = 291.2\ \ N\text{-}m$$

The Synchronous Generator operating alone: Example

The induced countertorque of the generator is

$$\tau_{app} = \frac{P_{conv}}{\omega_m} = \frac{34.1}{125.7} = 271.3 \ N\text{-}m$$

e. The voltage regulation of the generator is

Lagging PF:
$$VR = \frac{480 - 410}{410} \cdot 100\% = 17.1\%$$

Unity PF:
$$VR = \frac{480 - 468}{468} \cdot 100\% = 2.6\%$$

Lagging PF:
$$VR = \frac{480 - 535}{535} \cdot 100\% = -10.3\%$$

Terminal characteristics of Synchronous Generators

All generators are driven by a prime mover, such as a steam, gas, water, wind turbines, diesel engines, etc. Regardless the power source, most of prime movers tend to slow down with increasing the load. This decrease in speed is usually nonlinear but governor mechanisms of some type may be included to linearize this dependence.

The speed drop (SD) of a prime mover is defined as:

$$SD = \frac{n_{nl} - n_{fl}}{n_{fl}} \cdot 100\% \qquad (7.44.1)$$

Most prime movers have a speed drop from 2% to 4%. Most governors have a mechanism to adjust the turbine's no-load speed (set-point adjustment).

Frequency-Power Characteristics

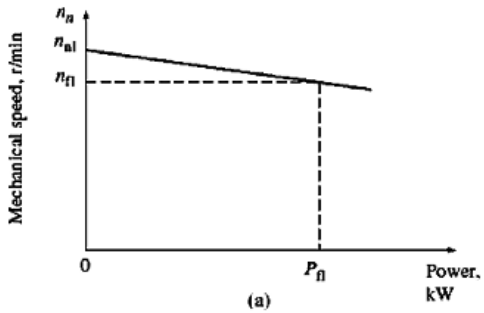
- The speed droop (**SD**) of a prime mover is defined by the following equation:

$$SD = \frac{n_{nl} - n_{fl}}{n_{fl}} x100\%$$

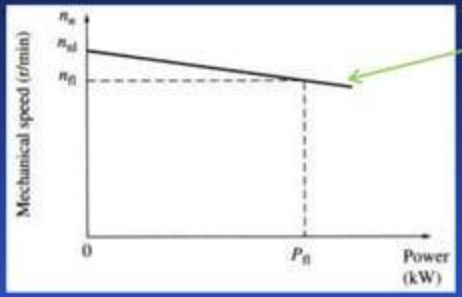
n_{nl} → No-load speed of the prime mover

n_{fl} → Full-load speed of the prime mover

- Most generator prime movers have a speed droop of **2 to 4 %**

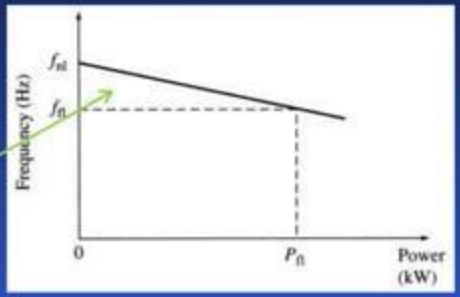


Terminal Characteristics of Synchronous Generators



A typical speed vs. power plot

A typical frequency vs. power plot



Since the shaft speed is linked to the electrical frequency as

$$f_e = \frac{n_m P}{120} \tag{7.45.1}$$

the power output from the generator is related to its frequency:

$$P = S_p \left(f_{nl} - f_{sys} \right) \tag{7.45.2}$$

Slope of curve, W/Hz

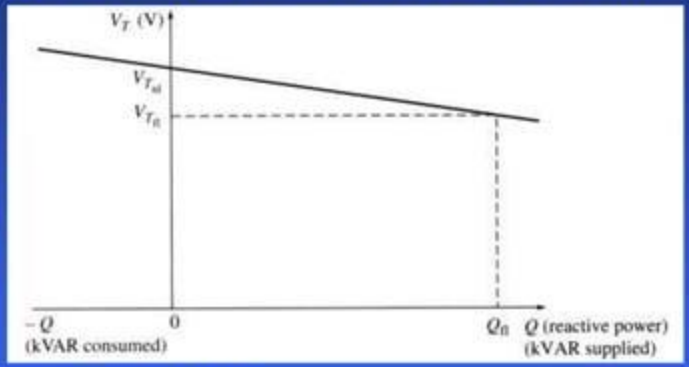
Operating frequency of the system

Terminal characteristics of Synchronous Generators

A similar relationship can be derived for the reactive power Q and terminal voltage V_T. When adding a lagging load to a synchronous generator, its terminal voltage decreases. When adding a leading load to a synchronous generator, its terminal voltage increases.

The plot of terminal voltage vs. reactive power is not necessarily linear.

Both the frequency-power and terminal voltage vs. reactive power characteristics are important for parallel operations of generators.

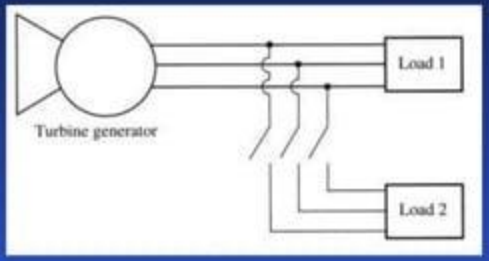


When a generator is operating alone supplying the load:

1. The real and reactive powers are the amounts demanded by the load.
2. The governor of the prime mover controls the operating frequency of the system.
3. The field current controls the terminal voltage of the power system.

Terminal characteristics of Synchronous Generators: Example

Example 7.3: A generator with no-load frequency of 61.0 Hz and a slope s_p of 1 MW/Hz is connected to Load 1 consuming 1 MW of real power at 0.8 PF lagging. Load 2 (that is to be connected to the generator) consumes a real power of 0.8 MW at 0.707 PF lagging.



a. Find the operating frequency of the system before the switch is closed.
b. Find the operating frequency of the system after the switch is closed.
c. What action could an operator take to restore the system frequency to 60 Hz after both loads are connected to the generator?

The power produced by the generator is

$$P = s_p\left(f_{nl} - f_{sys}\right)$$

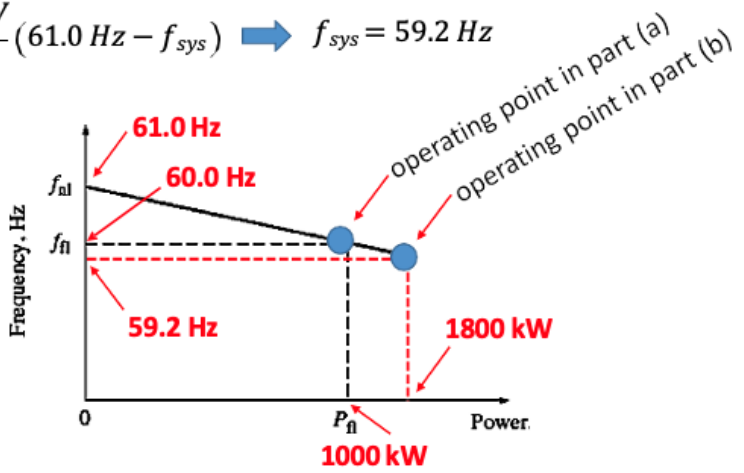
Therefore:

$$f_{sys} = f_{nl} - \frac{P}{s_p}$$

(b) After **Load 2** is connected, the operating frequency of the system is found as follows:

$$P = sp(f_{nl} - f_{sys})$$

$$(1000\,kW + 800\,kW) = \frac{1MW}{Hz}(61.0\,Hz - f_{sys}) \Rightarrow f_{sys} = 59.2\,Hz$$



As seen, **adding the load decreases the frequency** of the system from **60 Hz** to **59.2 Hz**

Terminal characteristics of Synchronous Generators: Example

a. The frequency of the system with one load is

$$f_{sys} = f_{nl} - \frac{P}{s_p} = 61 - \frac{1}{1} = 60\ Hz$$

b. The frequency of the system with two loads is

$$f_{sys} = f_{nl} - \frac{P}{s_p} = 61 - \frac{1.8}{1} = 59.2\ Hz$$

c. To restore the system to the proper operating frequency, the operator should increase the governor no-load set point by 0.8 Hz, to 61.8 Hz. This will restore the system frequency of 60 Hz.

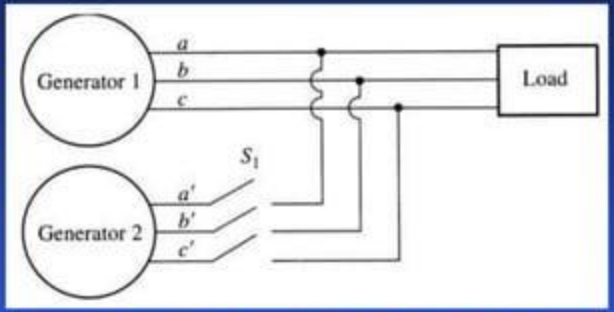
Parallel operation of Synchronous Generators

Most of synchronous generators are operating in parallel with other synchronous generators to supply power to the same power system. Obvious advantages of this arrangement are:

1. Several generators can supply a bigger load;
2. A failure of a single generator does not result in a total power loss to the load increasing reliability of the power system;
3. Individual generators may be removed from the power system for maintenance without shutting down the load;
4. A single generator not operating at near full load might be quite inefficient. While having several generators in parallel, it is possible to turn off some of them when operating the rest at near full-load condition.

Conditions required for paralleling

A diagram shows that Generator 2 (oncoming generator) will be connected in parallel when the switch S_1 is closed. However, closing the switch **at an arbitrary moment** can severely damage both generators!

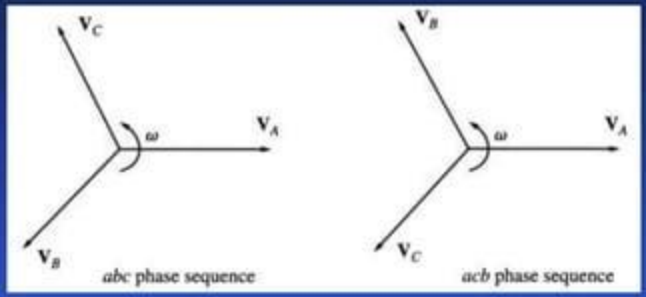


If voltages are not exactly the same in both lines (i.e. in *a* and *a'*, *b* and *b'* etc.), a very large current will flow when the switch is closed. Therefore, to avoid this, voltages coming from both generators must be exactly the same. Therefore, the following conditions must be met:

1. The rms line voltages of the two generators must be equal.
2. The two generators must have the same phase sequence.
3. The phase angles of two *a* phases must be equal.
4. The frequency of the oncoming generator must be slightly higher than the frequency of the running system.

Conditions required for paralleling

If the phase sequences are different, then even if one pair of voltages (phases *a*) are in phase, the other two pairs will be 120^0 out of phase creating huge currents in these phases.



If the frequencies of the generators are different, a large power transient may occur until the generators stabilize at a common frequency. The frequencies of two machines must be very close to each other but not exactly equal. If frequencies differ by a small amount, the phase angles of the oncoming generator will change slowly with respect to the phase angles of the running system.

If the angles between the voltages can be observed, it is possible to close the switch S_1 when the machines are in phase.

Conditions required for paralleling

- To check phase sequences we have three options:





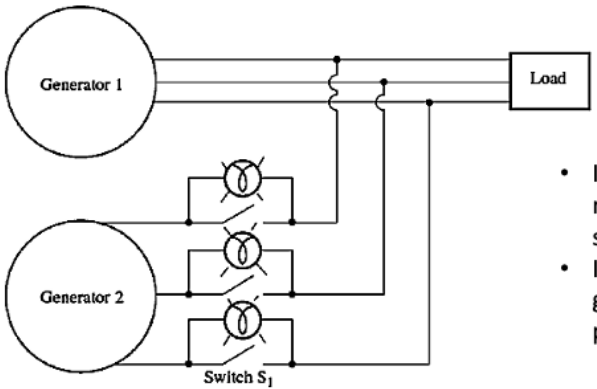
Option 1: use phase sequence indicator



Option 2: use a small three-phase induction motor and observe its rotation direction

Conditions required for paralleling

- To check phase sequences we have three options:

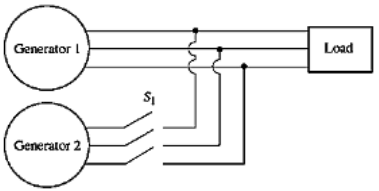


- If all three bulbs **get bright and dark together**, this means that both generators have the same phase sequence
- If the bulbs **brighten in succession**, this means that the generators have the opposite phase sequence, and two phases of the Gen 2 **must be exchanged**

Option 3: Use three-bulbs connected as shown

Conditions required for paralleling

3) The **frequency of the oncoming generator** must be adjusted to be **slightly higher** than the frequency of the running system. So that when it is connected, it will come on the line supplying power as a **generator**, instead of consuming the power as a **motor**



This check can be done by **two options**:

1) Use a frequency meter



2) Use a tachometer to measure the shaft speed of Gen 2

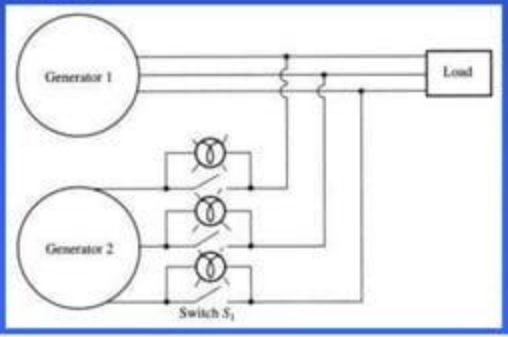


$$n_m = \frac{120 f_e}{P}$$

General procedure for paralleling generators

When connecting the generator G_2 to the running system, the following steps should be taken:

1. Adjust the field current of the oncoming generator to make its terminal voltage equal to the line voltage of the system (use a voltmeter).
2. Compare the phase sequences of the oncoming generator and the running system. This can be done by different ways:
   1) Connect a small induction motor to the terminals of the oncoming generator and then to the terminals of the running system. If the motor rotates in the same direction, the phase sequence is the same;
   2) Connect three light bulbs across the open terminals of the switch. As the phase changes between the two generators, light bulbs get brighter (large phase difference) or dimmer (small phase difference). If all three bulbs get bright and dark together, both generators have the same phase sequences.



Conditions required for paralleling

Option 2: Use a synchroscope



A ***synchroscope*** is a meter that measures the phase angle difference between the ***a-phases*** of the two systems.

How can we use synchronoscope ?

- The needle should **rotate slowly**.
- ***If it rotates to the right***: This means that the oncoming generator is **faster** than the running system (***the desired situation***)
- ***If it rotates to the left***: This means that the oncoming generator is **slower** than the running system
- When the synchroscope needle is **in the vertical position**, **we can close switch S1**

General procedure for paralleling generators

If phase sequences are different, two of the conductors on the oncoming generator must be reversed.

3. The frequency of the oncoming generator is adjusted to be slightly higher than the system's frequency.
4. Turn on the switch connecting G_2 to the system when phase angles are equal.

The simplest way to determine the moment when two generators are in phase is by observing the same three light bulbs. When all three lights go out, the voltage across them is zero and, therefore, machines are in phase.

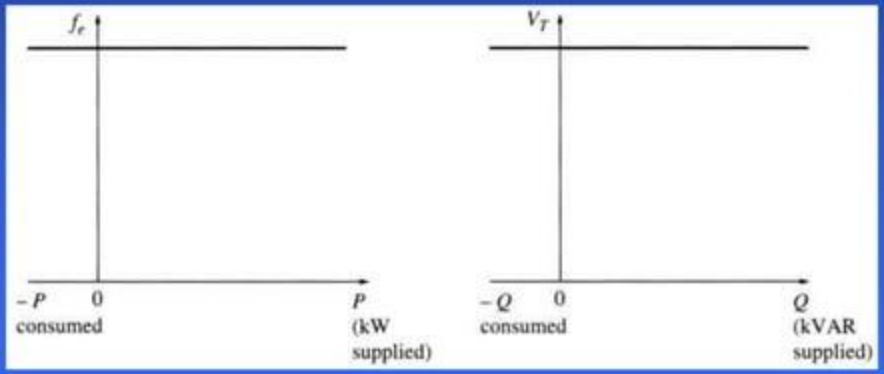
A more accurate way is to use a synchroscope – a meter measuring the difference in phase angles between two *a* phases. However, a synchroscope does not check the phase sequence since it only measures the phase difference in one phase.

The whole process is usually automated…



Operation of generators in parallel with large power systems

Often, when a synchronous generator is added to a power system, that system is so large that one additional generator does not cause observable changes to the system. A concept of an infinite bus is used to characterize such power systems. An infinite bus is a power system that is so large that its voltage and frequency do not vary regardless of how much real and reactive power is drawn from or supplied to it. The power-frequency and reactive power-voltage characteristics are:

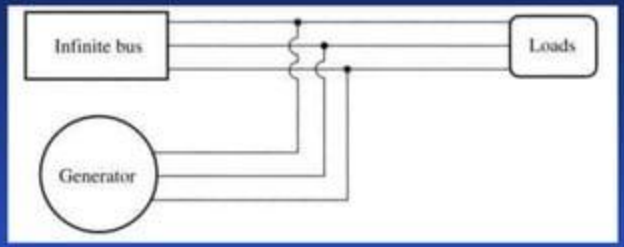


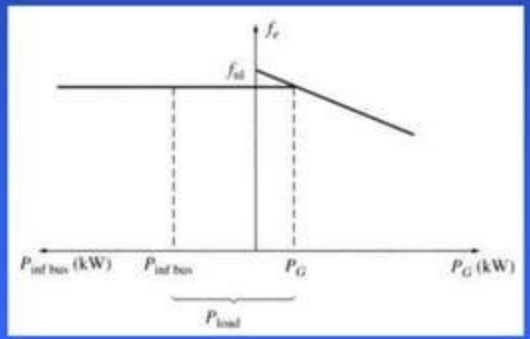
Operation of generators in parallel with large power systems

Consider adding a generator to an infinite bus supplying a load.

The frequency and terminal voltage of all machines must be the same. Therefore, their power-frequency and reactive power-voltage characteristics can be plotted with a common vertical axis.

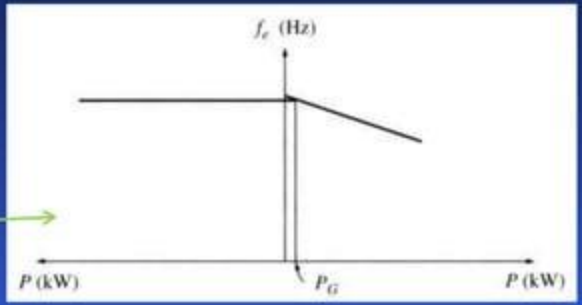
Such plots are called sometimes as house diagrams.



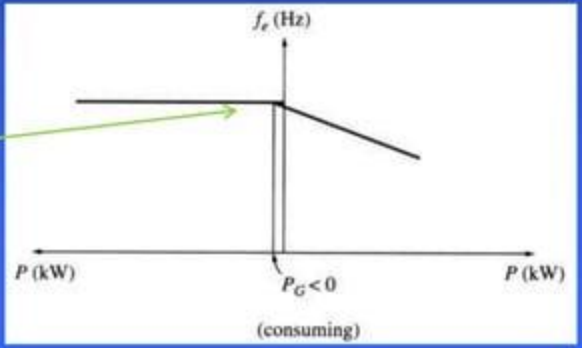


Operation of generators in parallel with large power systems

If the no-load frequency of the oncoming generator is slightly higher than the system's frequency, the generator will be "floating" on the line supplying a small amount of real power and little or no reactive power.

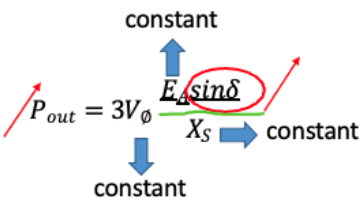


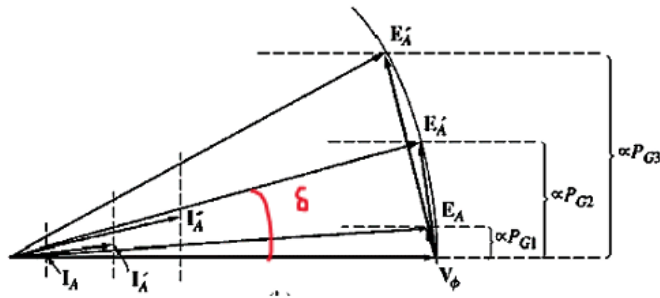
If the no-load frequency of the oncoming generator is slightly lower than the system's frequency, the generator will supply a negative power to the system: the generator actually consumes energy acting as a motor!
Many generators have circuitry automatically disconnecting them from the line when they start consuming energy.



Infinite bus

- Now, what happens to the **phasor diagram** of the generator if **its governor set-point** is **increased**?
- We know that (*from previous slide*) the **real power** supplied by the generator **increases**.





- According to this equation, **sinδ or torque angle "δ" must increase**

$$P_{G3} > P_{G2} > P_{G1}$$

$$\delta_3 > \delta_2 > \delta_1$$

Infinite bus

- What happens now, if the output of the generator is **further increased** so that it **exceeds** the power consumed by the load?

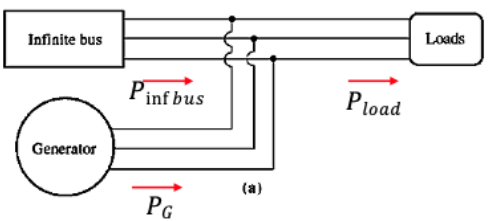
$P_G > P_{load}$

- If this occurs, the **extra power** generated by the generator **flows back into the infinite bus**.
- For example;

$P_{load} = 1000\ kW$
$P_G = 1300\ kW$

$P_{\inf bus} = -300\ kW$ (**generated**)

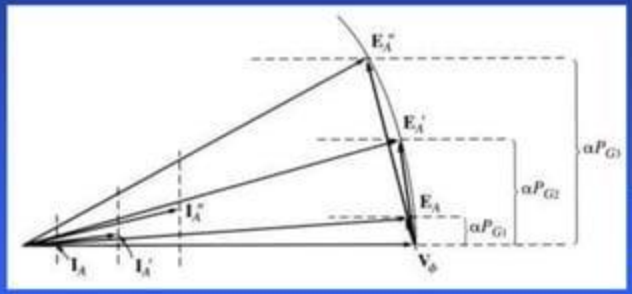
•Let's remember the infinite bus, by definition, the infinite can **supply** or **consume any amount of power** without **a change in frequency**, so the **extra power is consumed**.

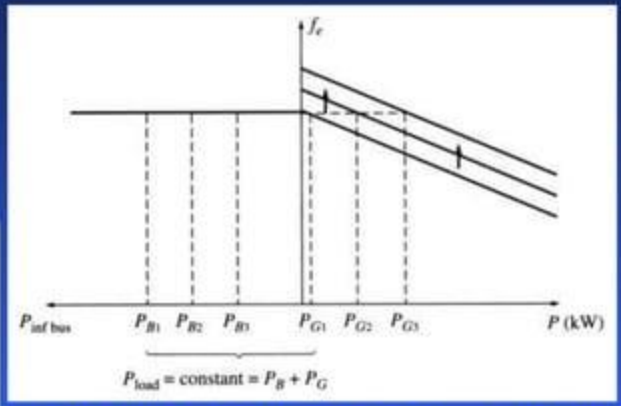


$P_{load} = P_{\inf bus} + P_G$

Operation of generators in parallel with large power systems

If the frequency of the generator is increased after it is connected to the infinite bus, the system frequency cannot change and the power supplied by the generator increases.





Notice that when E_A stays constant (field current and speed are the same), $E_A \sin\delta$ (which is proportional to the output power if V_T is constant) increases.

If the frequency of the generator is further increased, power output from the generator will be increased and at some point it may exceed the power consumed by the load. This extra power will be consumed by ~~the load~~.

Operation of generators in parallel with large power systems

After the real power of the generator is adjusted to the desired value, the generator will be operating at a slightly leading PF acting as a capacitor that consumes reactive power. Adjusting the field current of the machine, it is possible to make it to supply reactive power *Q* to the system.

Summarizing, when the generator is operating in parallel to an infinite bus:

1. The frequency and terminal voltage of the generator are controlled by the system to which it is connected.
2. The governor set points of the generator control the real power supplied by the generator to the system.
3. The generator's field current controls the reactive power supplied by the generator to the system.

Infinite bus

- So far we have seen that what happens to a synchronous generator connected to infinite bus when the **governor set-point** is **increased**?
- Now, the next question will be what happens to a synchronous generator connected to infinite bus if **its field current is increased? (governor set-point is unchanged)**

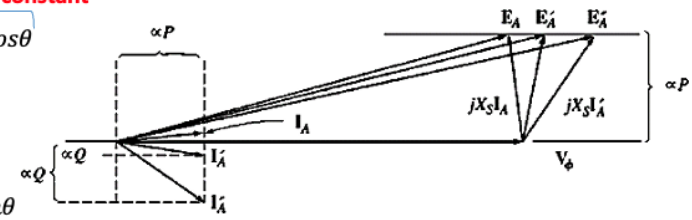
must be constant

$P_{out} = 3.V_{\emptyset}.I_A.cos\theta$

constant constant

$Q_{out} = 3.V_{\emptyset}.I_A.sin\theta$

constant increases



must be constant

$P_{out} = 3V_{\emptyset} \frac{E_A sin\delta}{X_S}$

constant constant constant

Infinite bus

- In summary, what happens to a synchronous generator connected to infinite bus if **its field current is increased? (governor set-point is unchanged)**

- The **real power output** of the generator **does not change** (*since governor set-point is unchanged*)
- Since **real power output** of the generator is **constant** $E_A sin\delta$ must be **constant**
- Since **real power output** of the generator is **constant** $I_A cos\theta$ must be **constant**
- The **magnitude** (rms) of the armature current $|I_A|$ **increases**
- Since $|I_A|$ **increases** and $I_A cos\theta$ is **constant,** then $I_A sin\theta$ must **increase**
- Since $I_A sin\theta$ **increases**, Q_{out} **increases** as well.

So, **as a result:**

- ***increasing the field current in a synchronous generator operating in parallel with an infinite bus increases the reactive power output of the generator***

Infinite bus

To summarize;

- **When a single generator is operating alone:**

1) The real and reactive powers supplied by the generator are fixed, and constrained to be equal to the power demanded by the load

2) The frequency and terminal voltage are varied by the governor set points and the field current, respectively.

- **When a generator is operating in parallel with an infinite bus:**

1) The frequency and terminal voltage of the generator are controlled by the system to which it is connected.

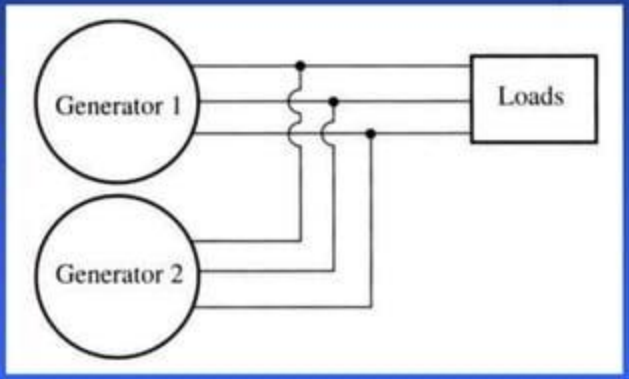
2) The governor set points of the generator control the real power supplied by the generator to the system.

3) The field current in the generator controls the reactive power supplied by the generator to the system.

Generators in parallel with other generators of the same size

When a generator is working alone, its real and reactive power are fixed and determined by the load.
When a generator is connected to an infinite bus, its frequency and the terminal voltage are constant and determined by a bus.

When two generators of the same size are connected to the same load, the sum of the real and reactive powers supplied by the two generators must equal the real and reactive powers demanded by the load:



$$P_{tot} = P_{load} = P_{G1} + P_{G2} \qquad (7.59.1)$$

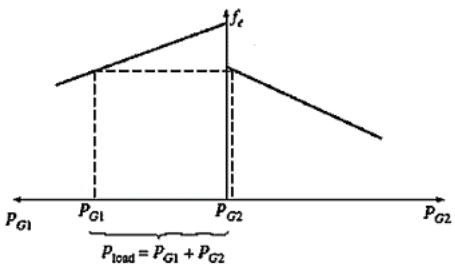
$$Q_{tot} = Q_{load} = Q_{G1} + Q_{G2} \qquad (7.59.2)$$

Parallel operation of generators with the same size

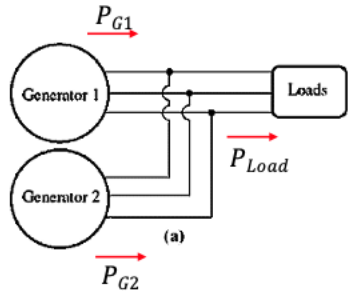
- If two generators are connected in parallel as shown in the figure, the real and reactive power of the load is shared between the generators.

$$P_{Load} = P_{G1} + P_{G2}$$

$$Q_{Load} = Q_{G1} + Q_{G2}$$

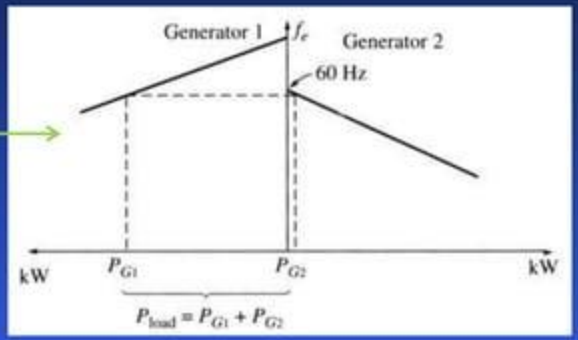


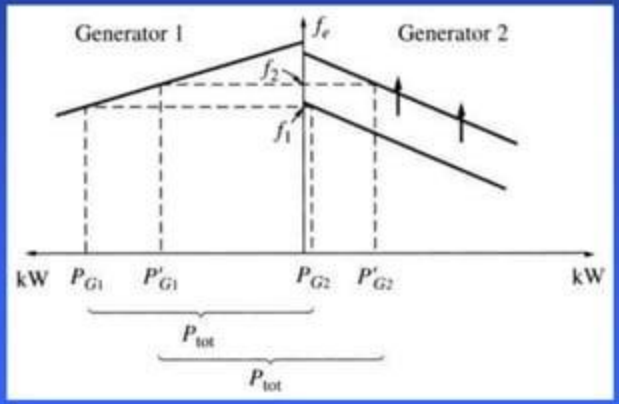
Two house diagrams shown together



Generators in parallel with other generators of the same size

Since the frequency of G_2 must be slightly higher than the system's frequency, the power-frequency diagram right after G_2 is connected to the system is shown.





If the frequency of G_2 is next increased, its power-frequency diagram shifts upwards. Since the total power supplied to the load is constant, G_2 starts supplying more power and G_1 starts supplying less power and the system's frequency increases.

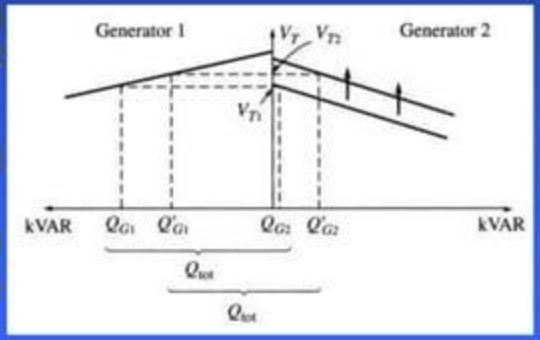
Generators in parallel with other generators of the same size

Therefore, when two generators are operating together, an increase in frequency (governor set point) on one of them:

1. Increases the system frequency.
2. Increases the real power supplied by that generator, while reducing the real power supplied by the other one.

When two generators are operating together, an increase in the field current on one of them:

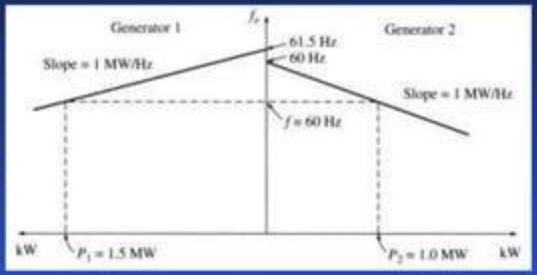
1. Increases the system terminal voltage.
2. Increases the reactive power supplied by that generator, while reducing the reactive power supplied by the other.



If the frequency-power curves of both generators are known, the powers supplied by each generator and the resulting system frequency can be determined.

Generators in parallel with other generators of the same size: Ex

Example 7.4: Two generators are set to supply the same load. Generator 1 has a no-load frequency of 61.5 Hz and a slope s_{p1} of 1 MW/Hz. Generator 2 has a no-load frequency of 61.0 Hz and a slope s_{p2} of 1 MW/Hz. The two generators are supplying a real load of 2.5 MW at 0.8 PF lagging.



a. Find the system frequency and power supplied by each generator.
b. Assuming that an additional 1 MW load is attached to the power system, find the new system frequency and powers supplied by each generator.
c. With the additional load attached (total load of 3.5 MW), find the system frequency and the generator powers, if the no-load frequency of G_2 is increased by 0.5 Hz.

The power produced by a synchronous generator with a given slope and a no-load frequency is

$$P = s_p \left(f_{nl} - f_{sys} \right)$$

Generators in parallel with other generators of the same size: Ex

The total power supplied by the generators equals to the power consumed by the load:

$$P_{load} = P_1 + P_2$$

a. The system frequency can be found from:

$$P_{load} = P_1 + P_2 = s_{p1}\left(f_{nl,1} - f_{sys}\right) + s_{p2}\left(f_{nl,2} - f_{sys}\right)$$

as $$f_{sys} = \frac{s_{p1}f_{nl,1} + s_{p2}f_{nl,2} - P_{load}}{s_{p1} + s_{p2}} = \frac{1 \cdot 61.5 + 1 \cdot 61.0 - 2.5}{1+1} = 60.0\ Hz$$

The powers supplied by each generator are:

$$P_1 = s_{p1}\left(f_{nl,1} - f_{sys}\right) = 1 \cdot \left(61.5 - 60\right) = 1.5\ MW$$

$$P_2 = s_{p2}\left(f_{nl,2} - f_{sys}\right) = 1 \cdot \left(61.0 - 60\right) = 1\ MW$$

Generators in parallel with other generators of the same size: Ex

b. For the new load of 3.5 MW, the system frequency is

$$f_{sys} = \frac{s_{p1} f_{nl,1} + s_{p2} f_{nl,2} - P_{load}}{s_{p1} + s_{p2}} = \frac{1 \cdot 61.5 + 1 \cdot 61.0 - 3.5}{1+1} = 59.5\ Hz$$

The powers are:

$$P_1 = s_{p1}\left(f_{nl,1} - f_{sys}\right) = 1 \cdot \left(61.5 - 59.5\right) = 2.0\ MW$$

$$P_2 = s_{p2}\left(f_{nl,2} - f_{sys}\right) = 1 \cdot \left(61.0 - 59.5\right) = 1.5\ MW$$

c. If the no-load frequency of G_2 increases, the system frequency is

$$f_{sys} = \frac{s_{p1} f_{nl,1} + s_{p2} f_{nl,2} - P_{load}}{s_{p1} + s_{p2}} = \frac{1 \cdot 61.5 + 1 \cdot 61.5 - 3.5}{1+1} = 59.75\ Hz$$

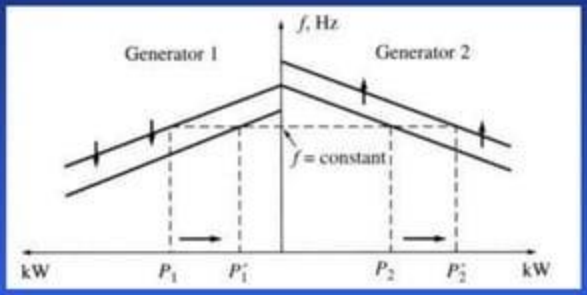
The powers are:

$$P_1 = P_2 = s_{p1}\left(f_{nl,1} - f_{sys}\right) = 1 \cdot \left(61.5 - 59.75\right) = 1.75\ MW$$

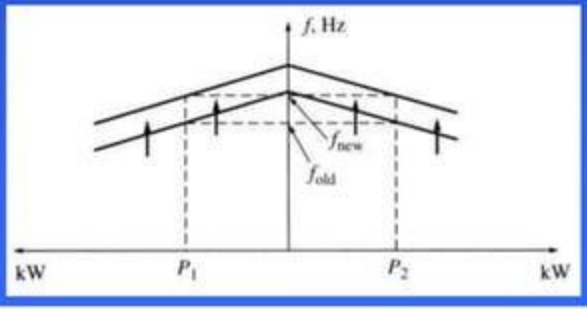
Generators in parallel with other generators of the same size

When two generators of the same size are working in parallel, a change in frequency (governor set points) of one of them changes both the system frequency and power supplied by each generator.

To adjust power sharing without changing the system frequency, we need to increase the frequency (governor set points) of one generator and simultaneously decrease the frequency of the other generator.

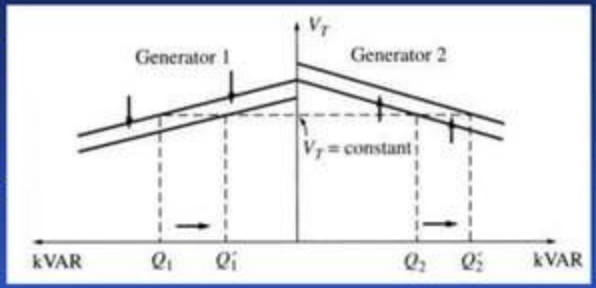


To adjust the system frequency without changing power sharing, we need to simultaneously increase or decrease the frequency (governor set points) of both generators.

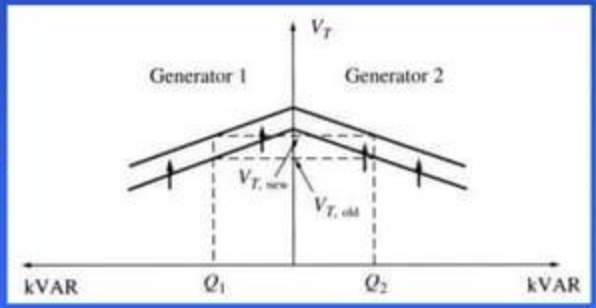


Generators in parallel with other generators of the same size

Similarly, to adjust the reactive power sharing without changing the terminal voltage, we need to increase simultaneously the field current of one generator and decrease the field current of the other generator.



To adjust the terminal voltage without changing the reactive power sharing, we need to simultaneously increase or decrease the field currents of both generators.



Generators in parallel with other generators of the same size

It is important that both generators being paralleled have dropping frequency-power characteristics.

If two generators have flat or almost flat frequency-power characteristics, the power sharing between them can vary widely with only finest changes in no-load speed. For good of power sharing between generators, they should have speed drops of 2% to 5%.

